

CORRELATION & T-TEST

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Loading the packages.

```
library(tidyverse)
library(janitor)
library(ggcorrplot)
library(corrplot)
library(readr)
```

Loading the dataset.

```
data <- read_csv("C:/Users/Nobert Wafula/Desktop/Diabetes_Dataset.csv")
```

Data Exploration.

```
glimpse(data)
```

```
## Rows: 50,000
## Columns: 8
## $ Glucose      <dbl> 103, 180, 165, 107, 90, 111, 76, 129, 112,
12...
## $ BloodPressure <dbl> 47, 72, 80, 60, 75, 75, 75, 60, 61, 58, 7
7, 6...
## $ SkinThickness <dbl> 35, 23, 25, 21, 28, 40, 20, 13, 31, 28, 3
6, 3...
## $ Insulin      <dbl> 149, 126, 154, 149, 68, 75, 136, 159, 100,
14...
## $ BMI          <dbl> 33.0, 33.4, 40.1, 17.4, 26.9, 27.8, 19.8,
37....
## $ DiabetesPedigreeFunction <dbl> 1.418, 2.274, 2.385, 1.101, 0.172, 0.882,
1.1...
## $ Age          <dbl> 42, 36, 59, 29, 42, 23, 35, 59, 45, 41, 3
2, 3...
## $ Outcome      <chr> "Negative", "Positive", "Positive", "Nega
tive..."
```

```
anyNA(data)
```

```
## [1] FALSE
```

```
sum(duplicated(data))
```

```
## [1] 0
```

Data Preparation.

```
data$Outcome <- as.factor(data$Outcome)
```

```
numeriv_vars <- data %>%
  select(where(is.numeric))
```

```
cor_matrix <- cor(numeriv_vars)
```

```
ggcorrplot(cor_matrix)
```

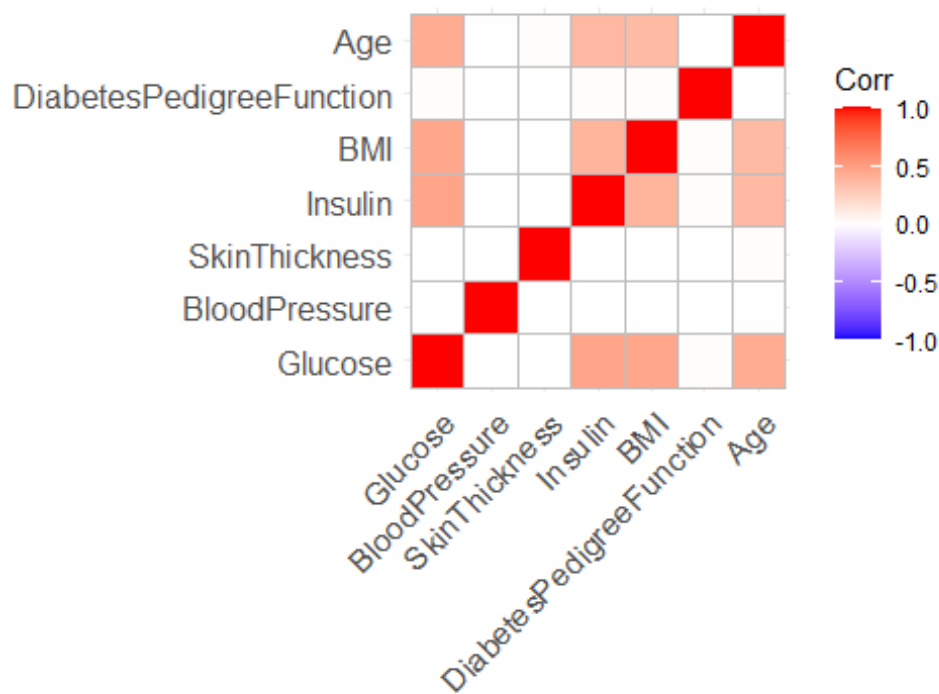
CORRELATION.

Pearson Correlation between Glucose and Insulin

```
cor1 = data %>% select(-Outcome) %>% cor(method = "pearson")  
cor1
```

##	Glucose	BloodPressure	SkinThickness	In
sulin				
## Glucose	1.000000000	-3.574832e-03	0.0024968778	4.73305
6e-01				
## BloodPressure	-0.003574832	1.000000e+00	0.0003981289	-3.90058
3e-06				
## SkinThickness	0.002496878	3.981289e-04	1.0000000000	4.39639
8e-03				
## Insulin	0.473305578	-3.900583e-06	0.0043963978	1.00000
0e+00				
## BMI	0.461195120	-7.167618e-03	0.0015759993	3.93486
1e-01				
## DiabetesPedigreeFunction	0.010518075	1.980186e-03	-0.0016370069	9.44238
6e-03				
## Age	0.434938287	-6.957472e-03	0.0076024704	3.70174
1e-01				
##	BMI	DiabetesPedigreeFunction	Ag	
e				
## Glucose	0.461195120	0.010518075	0.43493828	
7				
## BloodPressure	-0.007167618	0.001980186	-0.00695747	
2				
## SkinThickness	0.001575999	-0.001637007	0.00760247	
0				
## Insulin	0.393486128	0.009442386	0.37017405	
3				
## BMI	1.000000000	0.011203936	0.36065888	
0				
## DiabetesPedigreeFunction	0.011203936	1.000000000	0.00426985	
4				
## Age	0.360658880	0.004269854	1.00000000	
0				

```
ggcorrplot(cor1)
```



The results showed a moderate positive correlation between glucose and insulin levels ($r = 0.473$),

indicating that higher glucose levels were associated with higher insulin levels. Glucose also showed

a moderate positive correlation with BMI ($r = 0.461$) and age ($r = 0.435$), suggesting that increased body mass index

and older age were associated with higher glucose levels. Additionally, insulin and BMI demonstrated

a weak-to-moderate positive correlation ($r = 0.393$). Most other variables exhibited very weak or negligible correlations,

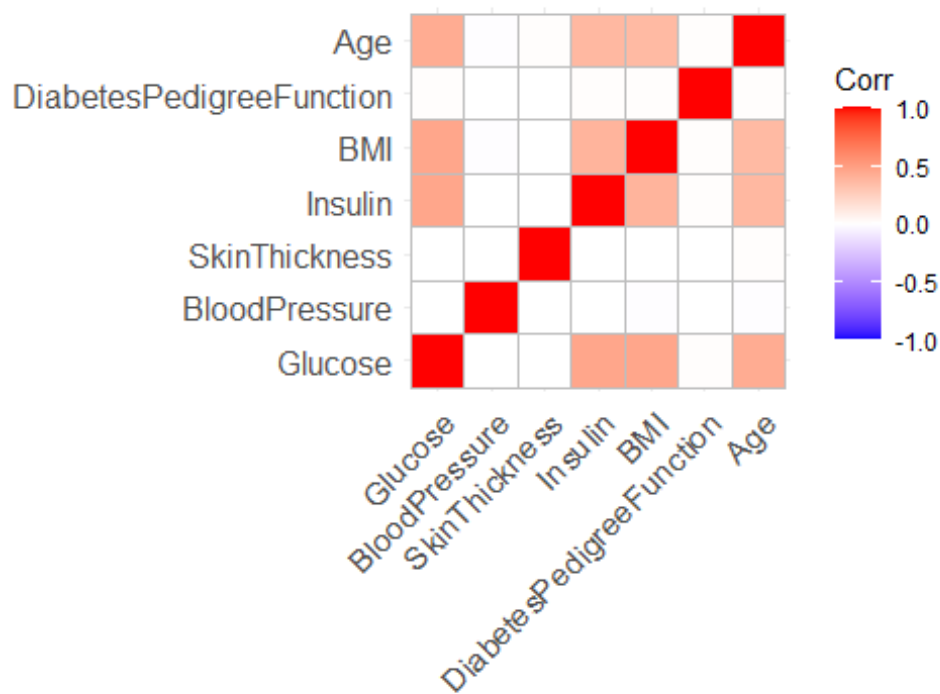
indicating minimal linear relationships among them.

Spearman Correlation

```
cor2 = data %>% select(-Outcome) %>% cor(method = "spearman")
cor2
```

```
##           Glucose BloodPressure SkinThickness
## Glucose      1.0000000000 -4.440542e-03 -3.454436e-04
## BloodPressure -0.0044405422  1.000000e+00  9.835103e-06
## SkinThickness -0.0003454436  9.835103e-06  1.000000e+00
## Insulin       0.4629853855 -6.366536e-04  2.329331e-04
## BMI           0.4628069689 -6.567119e-03  1.968561e-03
## DiabetesPedigreeFunction 0.0111575502  1.621933e-03 -3.123809e-03
## Age           0.4331850260 -5.108138e-03  6.617971e-03
##           Insulin           BMI DiabetesPedigreeFunction
on
## Glucose      0.4629853855  0.462806969           0.0111575
50
## BloodPressure -0.0006366536 -0.006567119           0.0016219
33
## SkinThickness 0.0002329331  0.001968561           -0.0031238
09
## Insulin       1.0000000000  0.386708101           0.0084760
49
## BMI           0.3867081011  1.000000000           0.0122475
29
## DiabetesPedigreeFunction 0.0084760487  0.012247529           1.0000000
00
## Age           0.3664224827  0.362831584           0.0050486
49
##           Age
## Glucose      0.433185026
## BloodPressure -0.005108138
## SkinThickness 0.006617971
## Insulin       0.366422483
## BMI           0.362831584
## DiabetesPedigreeFunction 0.005048649
## Age           1.000000000
```

```
ggcorrplot(cor2)
```



Spearman results are very similar to the Pearson results.

*This suggests:
the relationships are consistent,
and not heavily affected by outliers or non-normality.*

Kendall's tau correlation

```
cor3 <- cor(data$Glucose,
  data$BMI,
  method = "kendall")
```

```
cor3
```

```
## [1] 0.3110539
```

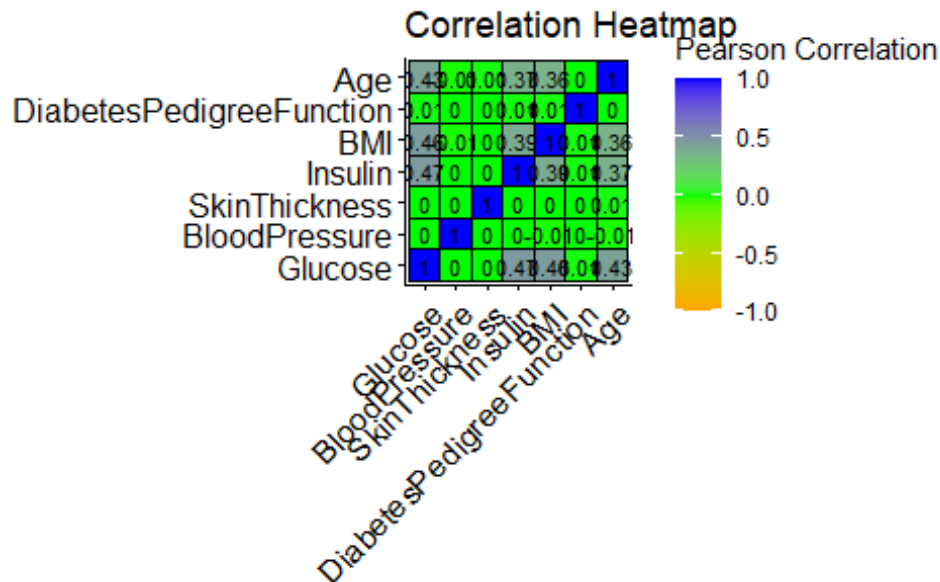
There is a relative moderate positive correlation between the glucose and BMI

This means that whenever glucose increases there is also a slight increase of BMI.

Visualization of the correlation matrix

```
ggcorrplot(cor1, title = "Correlation Heatmap", lab_col = "black",
  lab = TRUE, legend.title = "Pearson Correlation",
  lab_size = 3, ggtheme = theme_classic(),
```

```
outline.color = "black",
colors = c("orange", "green", "blue"))
```



One Sample T-Test

Test if average BMI differs from 25

```
t.test(data$BMI, mu = 25)
```

```
##
## One Sample t-test
##
## data: data$BMI
## t = 230.05, df = 49999, p-value < 2.2e-16
## alternative hypothesis: true mean is not equal to 25
## 95 percent confidence interval:
## 30.73020 30.82868
## sample estimates:
## mean of x
## 30.77944
```

A one-sample t-test I conducted to determine whether the mean BMI differed from 25.

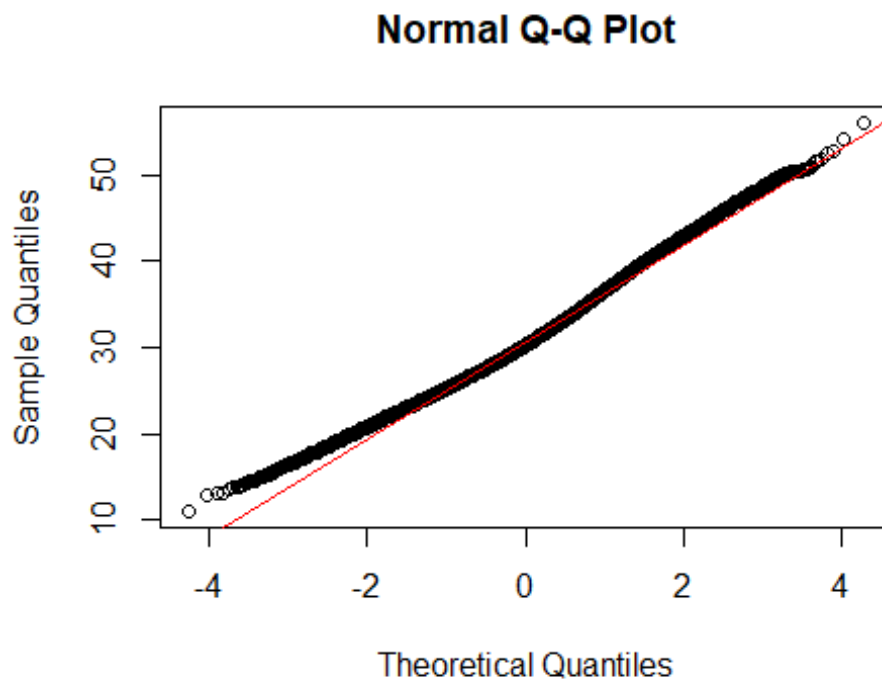
The results showed that the mean BMI ($M = 30.78$) is significantly higher than 25, $t(49999) = 230.05$, $p < 0.001$. The 95% confidence interval (30.73 to 30.83)

did not include 25, confirming a statistically significant difference. These results suggest that the average BMI in the dataset

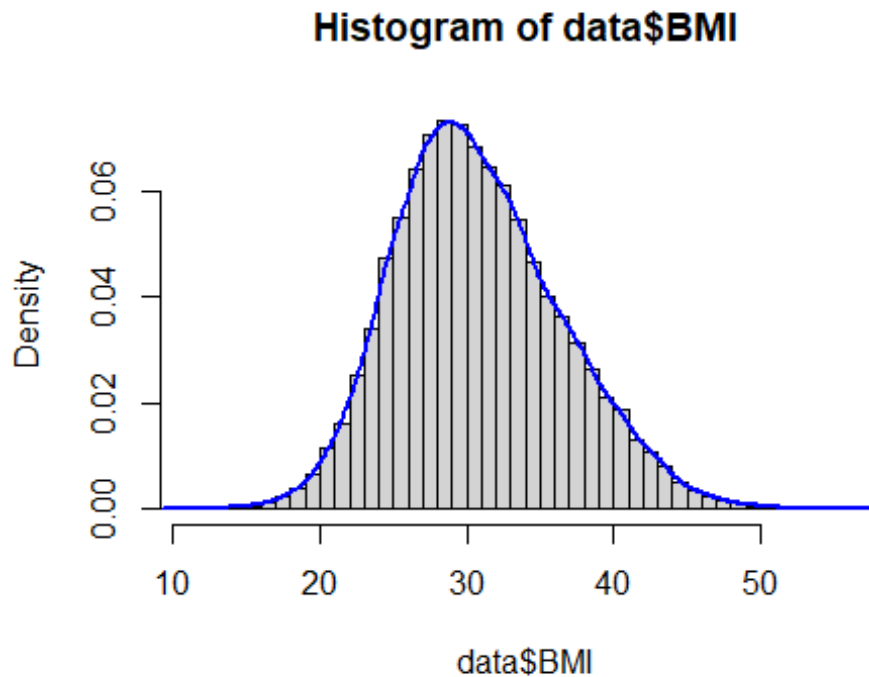
is significantly above the normal reference value of 25.
Therefore we reject the null hypothesis and conclude the alternative that the
observed mean is
Different from the theoretical one.

Checking for data normality.

```
qqnorm(data$BMI)  
qqline(data$BMI, col = "red")
```



```
hist(data$BMI, probability = TRUE, breaks = 50)  
lines(density(data$BMI), col = "blue", lwd = 2)
```



After i did visualizations the BMI is normally distributed.

Check for homogeneity of variance

```
library(car)
```

```
leveneTest(BMI ~ as.factor(Outcome), data = data)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
```

```
##           Df F value    Pr(>F)
```

```
## group      1 1114.2 < 2.2e-16 ***
```

```
##           49998
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

There is strong evidence that the variances of BMI are NOT equal across

Outcome groups.

This means you have heteroscedasticity (unequal variances).

Since the assumption is violated we need to use Wilcoxon Rank-Sum test.

Willcoxon test

```
wilcox.test(BMI ~ Outcome, data = data)
```

```
##
```

```
## Wilcoxon rank sum test with continuity correction
```

```
##
```

```
## data: BMI by Outcome
```



```
## W = 80493401, p-value < 2.2e-16
```

```
## alternative hypothesis: true location shift is not equal to 0
```

Null Hypothesis (H_0): The distribution of BMI is the same in both groups (Outcome = Negative and Outcome = Positive).

Alternative Hypothesis (H_1): The distribution (or median) of BMI is different between the two groups.

p-value < 2.2e-16 (essentially 0) → this is extremely significant.

Therefore, we reject the null hypothesis.

There is very strong evidence that BMI differs between the two Outcome groups.

👉 BMI is significantly associated with Outcome

Describe Direction of Difference

```
aggregate(BMI ~ Outcome, data = data, median)
```

```
##      Outcome      BMI
```

```
## 1 Negative 27.9
```

```
## 2 Positive 35.0
```

Individuals with no diabetes ("Negative") have a median BMI of 27.9.

Individuals with diabetes ("Positive") have a higher median BMI of 35.0.

Since the Wilcoxon test showed a highly significant p-value ($p < 2.2e-16$), and the median BMI is clearly higher in the Positive group, we can conclude:

There is a statistically significant difference in BMI between diabetic and non-diabetic individuals, with diabetic individuals having a higher median BMI.

Paired sample ttest

Create paired sample data

```
data_paired <- data.frame(
  ID = 1:10,
  BP_Before = c(135, 142, 130, 138, 140, 137, 145, 139, 136, 141),
  BP_After  = c(128, 134, 125, 130, 132, 129, 136, 130, 128, 133)
)
```

View the dataset

```
print(data_paired)
```

```
##      ID BP_Before BP_After
## 1     1      135      128
## 2     2      142      134
## 3     3      130      125
## 4     4      138      130
## 5     5      140      132
## 6     6      137      129
## 7     7      145      136
## 8     8      139      130
## 9     9      136      128
## 10    10      141      133
```

Perform paired t-test

```
t.test(data_paired$BP_Before, data_paired$BP_After, paired = TRUE)
```

```
##
## Paired t-test
##
## data: data_paired$BP_Before and data_paired$BP_After
## t = 21.726, df = 9, p-value = 4.369e-09
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
## 6.987861 8.612139
## sample estimates:
## mean difference
## 7.8
```

Check Assumptions

Check difference normality

```
diff_bp <- data_paired$BP_Before - data_paired$BP_After
shapiro.test(diff_bp) # Normality of difference scores
```

```
##
## Shapiro-Wilk normality test
##
## data: diff_bp
## W = 0.75164, p-value = 0.003737
```

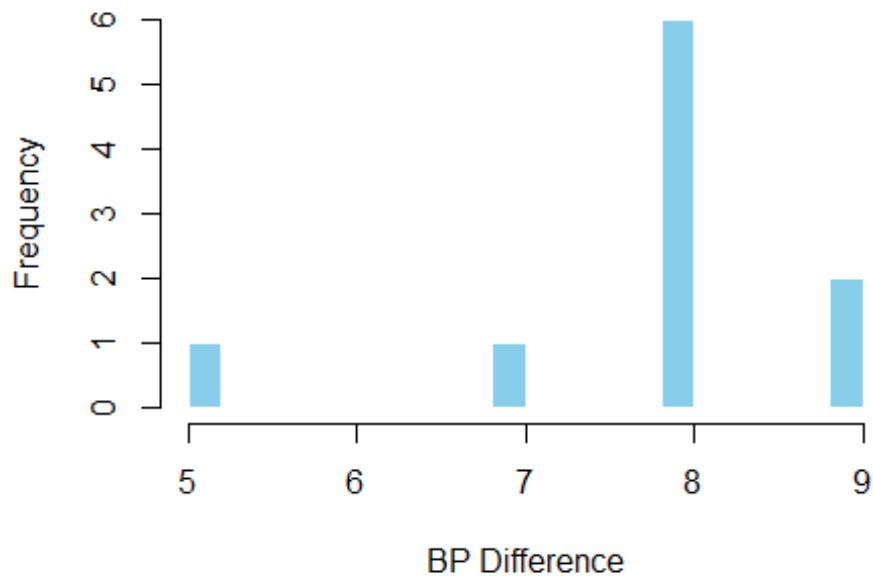
Since p-value = 0.003737 < 0.05, you reject the null hypothesis.

*This means that the **differences are not normally distributed, violating the normality assumption required for the paired t-test.*

Histogram.

```
hist(diff_bp,
      main = "Distribution of Blood Pressure Differences",
      xlab = "BP Difference",
      col = "skyblue",
      border = "white",
      breaks = 20)
```

Distribution of Blood Pressure Differences



Wilcoxon

```
wilcox.test(data_paired$BP_Before, data_paired$BP_After, paired = TRUE)
```

```
##
```

```
## Wilcoxon signed rank exact test
```

```
##
```

```
## data: data_paired$BP_Before and data_paired$BP_After
```

```
## V = 55, p-value = 0.001953
```

```
## alternative hypothesis: true location shift is not equal to 0
```

```
median(data_paired$BP_Before - data_paired$BP_After)
```

```
## [1] 8
```

Since the median is 8 positive it means that there was a decrease of median by 8 units after treatment.

There is statistically significant evidence that blood pressure decreased after the intervention, with a median reduction of approximately 8 units.